Aayushi Shrivastava

Education

Present	**Ph.D. in Mechanical Engineering (Controls)**, University of California Berkeley
May 2024	**Master of Science in Robotics**, University of Michigan Ann Arbor
May 2019	**Bachelor of Technology in Electrical Engineering**, Indian Institute of Technology (IIT) Roorkee, India

Publications

Sept 2026	**A. Shrivastava**, and et al. SAMBAR: Selective Anchoring via Method of Multipliers for Balanced Knowledge Acquisition and Retention in Vision-Language-Action Models. *(In Review)*. [arXiv]
Mar 2026	**A. Shrivastava**, K. Nagpal, S Jinkala, JB Bouvier, N. Mehr. Learning Control Policies to Provably Satisfy Hard Affine Constraints for Black-Box Hybrid Dynamical Systems. *Conference for Decision and Control. 2026 (Accepted)*. [arXiv]
Sept 2026	D. Dong, M. Bhatt, **A. Shrivastava**, and et al. Residual Denoising Enables Sample-Efficient Multi-Agent Coordination on Demand. *(In Review)*. [arXiv]
Sept 2026	S. Choi, S. Ye, N. Bui, **A. Shrivastava**, and et al. HuGo: LLMs as Whole-Body Policy Code Designers for Humanoid Loco-Manipulation. *(In Review)*. [arXiv]
Sept 2026	O. Dosunmu-Ogunbi, **A. Shrivastava**. Multi-Terrain Mastery: A Comprehensive Controller for Bipedal Locomotion. *International Conference on Electrical, Computer, Communications and Mechatronics Engineering (ICECCME)* 2026 *(Accepted)* [arXiv]
Oct 2024	O. Dosunmu-Ogunbi, **A. Shrivastava**, and et al. Demonstrating a Robust Walking Algorithm for Underactuated Bipedal Robots in Non-flat, Non-stationary Environments. *International Conference on Intelligent Robots and Systems*. 2024. [arXiv]
Oct 2023	O. Dosunmu-Ogunbi, **A. Shrivastava**, G. Gibson, J. Grizzle. Stair Climbing using the Angular Momentum Linear Inverted Pendulum Model and Model Predictive Control. *International Conference on Intelligent Robots and Systems*. 2023. [arXiv]

Research Experience

Present Sept 2026	**Making VLAs Robust using Method of Multipliers with Importance Weighting** The Berkeley Intelligent Control (ICON) Lab, *University of California, Berkeley* ○ Developing a novel learning framework to make Vision-Language-Action (VLA) models more robust by leveraging **Method of multipliers** and Importance Weighting. ○ Using a combination of Supervised Finetuning with Reinforcement Learning to adapt VLA policies against perturbations in initial object positions and camera positions.
Present Sept 2026	**Safe Whole Body Policies for the humanoid** The Berkeley Intelligent Control (ICON) Lab, *University of California, Berkeley* ○ Developing a novel continual learning framework for acquiring a safe whole-body loco-manipulation policy for humanoids. ○ Using Method of Multipliers and Importance Weighting to learn a safe policy without relying on prior knowledge of off-the-shelf controllers.
Sept 2026 Jan 2026	**SAMBAR: Replay Free Continual Learning Algorithm for VLAs** The Berkeley Intelligent Control (ICON) Lab, *University of California, Berkeley* ○ Formulated a replay-free continual learning framework for VLA fine-tuning by using the method of multipliers with selective parameter anchoring using importance weighting, which prevents forgetting. ○ The method demonstrated prior task's knowledge retention and new task's knowledge acquisition across the LIBERO benchmark and on hardware.
Dec 2025 Aug 2024	**Provably Satisfy Hard Affine Constraints for Black-Box Hybrid Dynamical Systems** The Berkeley Intelligent Control (ICON) Lab, *University of California, Berkeley* ○ Designed Reinforcement Learning (RL) algorithms with provable satisfy affine constraints for black-box hybrid dynamical systems. ○ Developed repulsive affine policy in state-space regions near constraints to ensure constraint-respecting trajectory generation, validated on paddle juggler and constrained pendulum environment.

Apr 2024 – Sept 2022 | **A Comprehensive Controller for Multi-Terrain Bipedal Locomotion**
Biped Walking Lab, *University of Michigan Ann Arbor*
○ Developed a unified controller capable of enabling bipedal robots to walk on diverse terrains (sand, gravel, rocks, wet slopes), improving real-world adaptability.
○ Extended controller capabilities beyond stairs and inclines to compliant and slippery surfaces through ankle motor strategies.

Work Experience

Aug 2024 – May 2024 | Software Engineering Intern
Motion Planning and Control Team, *Apptronik Inc.*
○ Developed and implemented an Angular Momentum-based Linear Inverted Pendulum (ALIP) footstep planner for the Apollo humanoid robot.
○ Achieved reduced drift and improved responsiveness to center-of-mass velocity changes. The controller was commercially deployed.

July 2022 – July 2019 | Software Engineer
Windows Performance Development Team, *Qualcomm India Pvt. Ltd.*
○ Analyzed and optimized the performance of benchmarks and applications on Qualcomm's ARM-based processors.
○ Developed a hashing algorithm micro-benchmark using Windows APIs to measure performance characteristics.

TEACHING EXPERIENCE

Dec 2025 – Aug 2025 | **Graduate Student Instructor**
Advanced Control System 1 (ME 232), *University of California Berkeley*
○ Facilitating weekly technical discussion sections for 50+ graduate students, focusing on **state-space representation, Lyapunov stability, and optimal control**.
○ Bridging theoretical control proofs with practical implementation by developing supplementary simulation tutorials for complex dynamical systems.

Apr 2019 – Jan 2019 | **Teaching Assistant**
Artificial Neural Networks (EEN-351), *IIT Roorkee*
○ Developed Python-based curricula for 80+ students, implementing fundamental algorithms including **Backpropagation, CNNs, and RNNs** from scratch.
○ Conducted focused review sessions to simplify high-dimensional calculus and optimization concepts for undergraduate learners.

Leadership and Outreach Experience

- **Mentor**, MEng Capstone Project on quadruped locomotion using the Go2 robot, *University of California, Berkeley* (Present).
- **Co-Director**, Women+ in Robotics and Engineering (WiRE+), University of Michigan, Ann Arbor (2024).
- **Founder**, Zero-Gravity Group under the Lean-In Program, promoting peer mentorship for women at IIT Roorkee (2018–19).
- **First Female Secretary** in the 58-year history of Models and Robotics Section, Student Technical Council, IIT Roorkee (2018–19).

Awards

- Awarded the **David & Patricia P. Bogy Fellowship** (Department of Mechanical Engineering, UC Berkeley 2026) in recognition of distinguished academic record and promise for continued noteworthy achievement.
- Awarded the **Kedarnath Agarwala I.S.E. Memorial Trophy** (2019) for academic excellence and outstanding contribution to social service.
- Received the **IITR Annual Excellency Award** (2016) for exceptional academic, co-curricular, and extracurricular achievements.
- Awarded **Best Project Award** (2019) for Biped Robot in the Electrical Engineering department.

Media

- Work featured in IEEE Spectrum's **Video Friday** (April 5, 2024): [link]
- Work showcased in IEEE Spectrum's **Video Friday** (November 24, 2023): [link]